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KhetAI: Testing an Irrigation Advisory

Course: Software Testing and Quality Assurance (STQA) | Lab CA : Poster Presentation

Swadha
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Class: LY B.Tech IT | Div: A | Batch: ____

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1. INTRODUCTION

- System under test: KhetAI, a sugarcane irrigation advisory. It joins plot data, soil telemetry, and an Open-Meteo forecast in an FAO-56 engine.
- A wrong badge can tell a farmer to irrigate when the reading is missing, illegal, or when rain is already expected.
- The decision is computed, so IRRIGATE_NOW, MONITOR, NO_IRRIGATION, and INSUFFICIENT_DATA can be asserted, not only looked at.
- The public site is khet-ai-irrigation-advisory.vercel.app. The API is the Express service the frontend rewrites to Render.

2. PROBLEM STATEMENT

- The advisory is unsafe if bad moisture, a stale sensor, or a null field still produces a pump recommendation.
- Defects to catch: humidity outside range accepted, empty registration, a plot opened without a token, and a unit mismatch treated as valid.
- Scope: the FAO-56 engine, sensor checks, and the auth API. Telemetry is marked SIMULATED. This is not a commercial field trial.

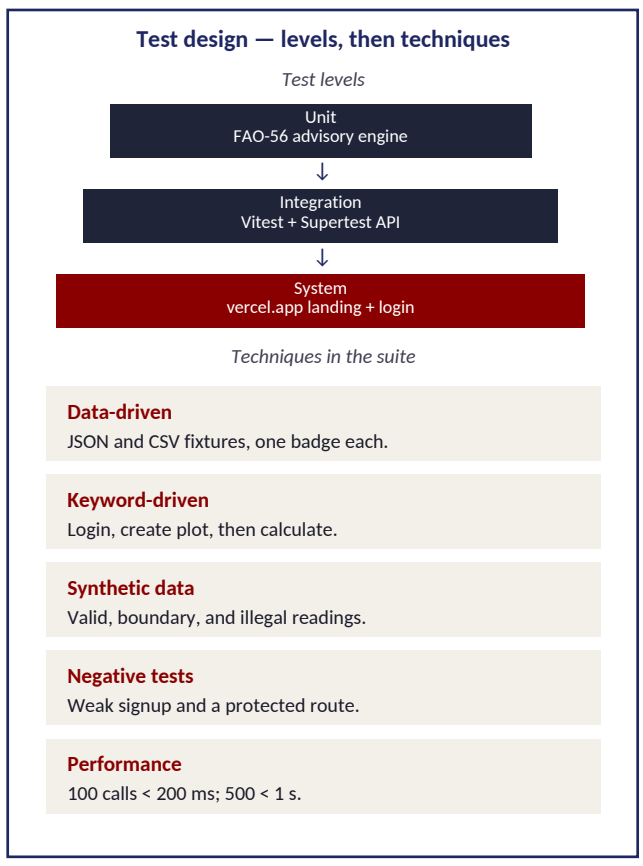
3. OBJECTIVES

- Match advisory badges to fixtures: IRRIGATE_NOW, IRRIGATE_SOON, MONITOR, NO_IRRIGATION, and INSUFFICIENT_DATA.
- Check boundaries: 0% and 100% soil moisture, out-of-range humidity, null input, and special characters.
- Check the API: weak registration is rejected, and a protected plot requires a token.
- Keep the experimental ML layer out of the pass/fail oracle. It is marked not directly comparable to the FAO-56 volume.

4. LITERATURE REVIEW

- Strategy: risk-based testing. A wrong irrigation call costs water and yield, so the decision badge is tested before the page polish.
- ISTQB Foundation Level — unit, integration, and system levels, plus data-driven and keyword-driven techniques.
- Myers, Sandler and Badgett — test the boundaries. Here those are 0% moisture, 100% moisture, and humidity outside the farm range.
- Sommerville — verification asks whether the FAO-56 rules hold; validation asks whether a farmer can trust the sentence on screen.
- Gap: the default run skips the PostgreSQL suite, and coverage is not measured. A green UI is not the same as a tested rule.

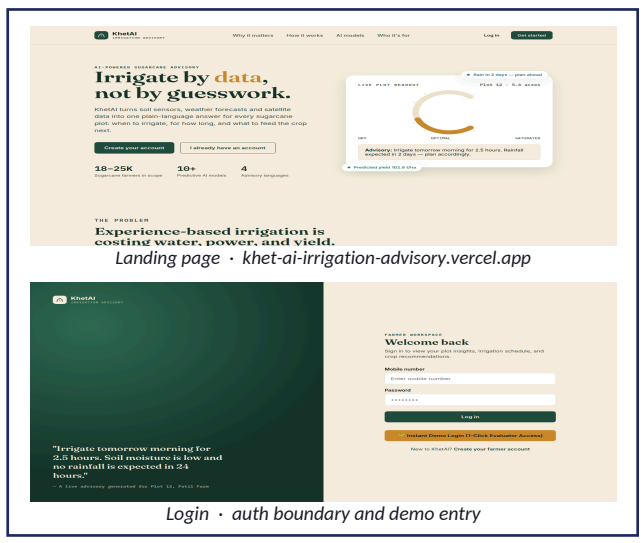
5. METHODOLOGY / SYSTEM DESIGN



6. IMPLEMENTATION DETAILS

- Tooling. Vitest 5 and Supertest, started with npm test in backend/. PostgreSQL tests are a separate opt-in file and need DATABASE_URL_TEST.
- Data-driven. Ten synthetic cases, TC_SYNTH_001 to 010, assert the badge. Null and special-character input must return INSUFFICIENT_DATA, not a pump time.
- Keyword-driven. Three scenarios chain LoginUser, CreatePlot, CalculateAdvisory, and ValidateSensorData. The invalid case rejects humidity outside the agricultural range.
- Result recorded in the repo: 71 tests passed, 0 failed, 7 skipped. The skipped cases are the live-database integration tests.

7. RESULTS & DISCUSSION



Landing page · khet-ai-irrigation-advisory.vercel.app

Login · auth boundary and demo entry

8. KEY FINDINGS

- The badge fixtures match, including INSUFFICIENT_DATA when the reading is null or not a number.
- Boundary moisture is classified. Humidity outside the farm range is rejected instead of stored.
- Weak registration fails, and a plot route without a token does not open another farmer's data.
- 71 passing tests still leave the database suite skipped, so ownership isolation is not in the default run.

9. CONCLUSION

- The FAO-56 advisory is testable. Badges, boundaries, and the auth API each have a passing case.
- Quality here means the app does not tell a farmer to run the pump on missing or illegal sensor data.
- The live landing page and login are the system surface. The suite is the evidence behind the recommendation.

10. FUTURE SCOPE

- Run the skipped PostgreSQL tests on a local khetai_test database, including cross-user access and teardown.
- Add a coverage gate, frontend checks for unsafe HTML, and weather timeout plus fallback tests.
- Limitation: soil telemetry is simulated, and the hosted demo-seed call currently returns HTTP 404.

11. REFERENCES

- [1] ISTQB. Certified Tester Foundation Level Syllabus. Levels and test techniques.
- [2] Myers, G.J., Sandler, C. and Badgett, T. The Art of Software Testing. Wiley.
- [3] Sommerville, I. Software Engineering. Pearson. Verification and validation.
- [4] ISO/IEC/IEEE 29119. Software testing — concepts and definitions.
- [5] KhetAI team. TESTING.md and backend/tests. Vitest suite, 71 passed, 7 skipped.

ACKNOWLEDGEMENT

- Faculty in-charge: Prof. Poonam Narkhede and Dr. Ankita Nagmothe.
- Project repository: github.com/Swadhaaaa/KhetAI-Irrigation-Advisory.